STA301 FINAL TERM CURRENT PAPRER 2022 SOLVED BY PRINCE ALVI

1) 2 q's means or variance k thy.

Variance = $\sigma 2 = E(X2) - [E(X)]2$

Variance of X = σ2 = E[(X - μ)2] = ∑(x - μ)2 P(x)

MEAN AND VARIANCE OF THE POPULATION DISTRIBUTION

Mean = $\mu = E(X) = \sum x f(x) = 2$

Variance = $\sigma^2 = Var(X) = E(X)^2 - [E(X)]^2 = \sum x^2 f(x) - [\sum x f(x)]^2$

2) Sampling distribution = $\hat{p} = \frac{X}{n}$

3) Normal distribution = $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2}$

4) Ak integral of function Ka tha. Jo prove krna tha.

EXAMPLE

Find the expected value of the random variable X having the p.d.f

f(x) = 2 (1 – x), 0 < x < 1

= 0, elsewhere

SOLUTION

Now $E(X) = \int_{-\infty}^{\infty} x f(x)\, dx$

$= 2\int_{0}^{1} x(1-x)\, dx$

$= 2\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = 2\left[\frac{1}{2} - \frac{1}{3}\right] = \frac{1}{3}$

5) one tailed

A test, for which the entire rejection region lies in only one of the two tails – either in the right tail or in the left tail – of the sampling distribution of the test-statistic, is called a one-tailed test or one-sided test.

H0 : p < 0.50 --- HA : p > 0.50 --- (one-tailed test)

6) hyper geometric = $P(X = x) = \frac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}}$

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EXAMPLE

The names of 5 men and 5 women are written on slips of paper and placed in a hat. Four names are drawn. What is the probability that 2 are men and 2 are women? Let us regard 'men' as success. Then X will denote the number of men. We have N = 5 + 5 = 10 names to be drawn from; Also, n = 4, (since we are drawing a sample of size 4 out of a 'population' of size 10) In addition, k = 5 (since there are 5 men in the population of 10). In this problem, the possible values of X are 0, 1, 2, 3, 4, i.e. n): The hyper geometric distribution is given by

$$P(X=x)=\frac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}},$$

Since N = 10, k = 5 and n = 4, hence, in this problem, the hyper geometric distribution is given by

$$P(X=x)=\frac{\binom{5}{x}\binom{5}{4-x}}{\binom{10}{4}}$$

and the required probability,
i.e P(X = 2) is

$$P(X=2)=\frac{\binom{5}{2}\binom{5}{4-2}}{\binom{10}{4}}$$

$$=\frac{\binom{5}{2}\binom{5}{2}}{\binom{10}{4}}$$

$$=\frac{10\times 10}{210}$$

$$=\frac{10}{21}$$

7) 95% confidence interval for μ : = $\bar{X}-1.96\frac{s}{\sqrt{n}}$ and $\bar{X}+1.96\frac{s}{\sqrt{n}}$, $s^2=\frac{\sum(X-\bar{X})^2}{n-1}$

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EXAMPLE-1

Consider a car assembly plant employing something over 25,000 men. In planning its future labour requirements, the management wants an estimate of the number of days lost per man each year due to illness or absenteeism.
A random sample of 500 employment records shows the following situation:

Number of Days Lost	Number of Employees
None	48
1 or 2	43
3 or 4	90
5 or 6	186
7 or 8	78
9 to 12	34
13 to 20	21
Total	500

Construct a 95% confidence interval for the mean number of days lost per man each year due to illness or absenteeism.

SOLUTION

- The point estimate of μ is $\bar{X}$, which in this example comes out to be $\bar{X} = 5.38$ days
- In order to construct a confidence interval for μ, we need to compute s, which in this example comes out to be $s = 3.53$ days.

Hence, the 95% confidence interval for μ comes out to be

$$\left(5.38 - \frac{1.96 \times 3.53}{\sqrt{500}}, 5.38 + \frac{1.96 \times 3.53}{\sqrt{500}}\right)$$

or 5.38 ± 0.31 days
= 5.07 days to 5.69 days.

8) Natural paired observation = $t = \dfrac{\bar{d}}{s_d/\sqrt{n}}$

9) Samples ki outcomes likni thi

Sample No.	Sample Data	Sample Proportion ($\hat{p}$)
1	1, 3, 6	1/3
2	1, 3, 8	1/3
3	1, 3, 9	0
4	1, 3, 12	1/3
5	1, 6, 8	2/3
6	1, 6, 9	1/3
7	1, 6, 12	2/3
8	1, 8, 9	1/3
9	1, 8, 12	2/3
10	1, 9, 12	1/3
11	3, 6, 8	2/3
12	3, 6, 9	1/3
13	3, 6, 12	2/3
14	3, 8, 9	1/3
15	3, 8, 12	2/3
16	3, 9, 12	1/3
17	6, 8, 9	2/3
18	6, 8, 12	1
19	6, 9, 12	2/3
20	8, 9, 12	2/3

10) chi- square distribution = $$f(x)=\frac{1}{2^{\nu/2}\Gamma(\nu/2)}(x)^{(\nu/2)-1}.e^{-x/2},\ 0<x<\infty$$

11) Significance level = We set the significance level at $\alpha = 0.05$.

12) Variance E(Y) = Var(Y) = E(Y2) – [E(Y)]2

13) critical region or level of signifinacnac ko compare krna tha
The level of significance acts as a basis for determining the CRITICAL REGION of the test. For example, if we are testing H0: $\mu = 45$ against H1: $\mu \neq 45$, our test statistic is the standard normal variable Z, and the level of significance is 5%, then the critical values are Z = +1.96
Corresponding to a level of significance of 5%, we have:

14) Expected value find krni the E(X),E(Y),E(XY) or Cov(X,Y) find krna tha

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EXAMPLE

From the following joint p.d. of X and Y, find Var(X), Var(Y), Cov(X,Y) and ρ.

x \ y	0	1	2	3	g(x)
0	0.05	0.05	0.10	0	0.20
1	0.05	0.10	0.25	0.10	0.50
2	0	0.15	0.10	0.05	0.30
h(y)	0.10	0.30	0.45	0.15	1.00

Now

E(X) $= \Sigma x_i g(x_i)$
$= 0 \times 0.20 + 1 \times 0.50 + 2 \times 0.30$
$= 0 + 0.50 + 0.60 = 1.10$

E(Y) $= \Sigma y_j h(y_j)$
$= 0 \times 0.10 + 1 \times 0.30 + 2 \times 0.45 + 3 \times 0.15$
$= 0 + 0.30 + 0.90 + 0.45 = 1.65$

E(X2) $= \Sigma x_i^2 g(x_i)$
$= 0 \times 0.20 + 1 \times 0.50 + 4 \times 0.30$
$= 1.70$

E(Y2) $= \Sigma y_j^2 h(y_j)$
$= 0 \times 0.10 + 1 \times 0.30 + 4 \times 0.45 + 9 \times 0.15$
$= 3.45$

Thus

Var(X) = E(X2) – [E(X)]2
$= 1.70 - (1.10)2 = 0.49,$

and

Var(Y) = E(Y2) – [E(Y)]2
$= 3.45 - (1.65)2 = 0.7275$

Again:

E(XY) $= \sum_i \sum_j (x_i y_j) f(x_i, y_j)$
$= 1 \times 0.10 + 2 \times 0.15 + 2 \times 0.25 + 4 \times 0.10 + 3 \times 0.10 + 6 \times 0.05$
$= 0.10 + 0.30 + 0.50 + 0.40 + 0.30 + 0.30$
$= 1.90$

$\therefore$ Cov(X,Y) = E(XY) – E(X) E(Y)
$= 1.90 - 1.10 \times 1.65 = 0.085,$ and

$$\rho = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)\,\text{Var}(Y)}}$$

$$= \frac{0.085}{\sqrt{(0.49)(0.7275)}} = \frac{0.085}{0.595}$$

$$= 0.14$$

Hence, we can say that there is a weak positive linear correlation between the random variables X and Y.

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15) CORRELATION CO-EFFICIENT = $\rho = \frac{Cov(X,Y)}{\sqrt{Var(X)\,Var(Y)}}$

16) t-distribution find krni thi

$$f(x) = \frac{1}{\sqrt{v}\,\beta\left(\frac{1}{2},\frac{v}{2}\right)}\left(1+\frac{x^2}{v}\right)^{-(v+1)/2}$$

17) Hypothesis null or alternative wala qustion tha ..

A null hypothesis should always be precise such as 'the given coin is unbiased' or 'a drug is ineffective in curing a particular disease' or 'there is no difference between the two teaching methods'.

An alternative hypothesis is any other hypothesis which we are willing to accept when the null hypothesis H0 is rejected. It is customarily denoted by H1 or HA. A null hypothesis H0 is thus tested against an alternative hypothesis H1. For example, if our null hypothesis is H0 : $\mu = 62''$, then our alternative hypothesis may be H1 : $\mu \neq 62''$ or H1 : $\mu < 62''$

18) Poisson distribution 3 properties

PROPERTIES OF THE POISSON DISTRIBUTION Some of the main properties of the Poisson distribution are given below:

- If the random variable X has a Poisson distribution with parameter μ, then its mean and variance are given by $E(X) = \mu$ and $Var(X) = \mu$.
- (In other words, we can say that the mean of the Poisson distribution is equal to its variance.)
- The shape of the Poisson distribution is positively skewed. The distribution tends to be symmetrical as μ becomes larger and larger.

19) 1 binomial k through X k value find krni thi. P and q k values given thin.

$$P(X = x) = \binom{n}{x} p^x q^{n-x}$$

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EXAMPLE

Suppose that we toss a fair coin 5 times, and we are interested in determining the probability distribution of X, where X represents the number of heads that we obtain. We note that in tossing a fair coin 5 times:

- Every toss results in either a head or a tail,
- The probability of heads (denoted by p) is equal to ½ every time (in other words, the probability of heads remains *constant*),
- Every throw is *independent* of every other throw, and
- The total number of tosses i.e. 5 is *fixed* in *advance*.

The above four points represents the *four basic* and vitally important *PROPERTIES* of binomial experiment. Now, in 5 tosses of the coin, there can be 0, 1, 2, 3, 4 or 5 heads, and the no. of heads is thus a random variable which can take one of these six values. In order to compute the probabilities of these X-values, the formula is:

Binomial Distribution

$$P(X=x)=\binom{n}{x}p^x q^{n-x}$$

Where
n = the total no. of trials
p = probability of success in each trial
q = probability of failure in each trial (i.e. q = 1 - p)
x = no. of successes in n trials.
x = 0, 1, 2, … n

The binomial distribution has two parameters, n and p. In this example, n = 5 since the coin was thrown 5 times, p = ½ since it is a fair coin, q = 1 – p = 1 – ½ = ½ Hence

Putting x = 0

$$P(X=x)=\binom{5}{x}\left(\tfrac{1}{2}\right)^x\ \left(\tfrac{1}{2}\right)^{5-x}$$

$$P(X=0)=\binom{5}{0}\left(\tfrac{1}{2}\right)^0\ \left(\tfrac{1}{2}\right)^{5-0}$$

$$=\frac{5!}{0!\,5!}(1)\left(\tfrac{1}{2}\right)^5$$

$$=1(1)\left(\tfrac{1}{2}\right)^5=\frac{1}{32}$$

Putting x = 1

20) Why t-distribution is a whole family of distributions?

The number of degrees of freedom determines the shape of the t-distribution. Thus there is a different t-distribution for each number of degrees of freedom. As such, it is a whole family of distributions.

21) If Population proportion (P) is 0.7, calculate mean and standard error for the sampling distribution of sample proportions ( ), when the sample size is 36(with replacement)?

a)

$$\sigma_{\hat{p}}=\sqrt{\frac{pq}{n}},\quad \sigma_{\hat{p}_1-\hat{p}_2}=\sqrt{\frac{p_1q_1}{n_1}+\frac{p_2q_2}{n_2}}$$

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EXAMPLE-2

Ten percent of the 1-kilogram boxes of sugar in a large warehouse are underweight. Suppose a retailer buys a random sample of 144 of these boxes. What is the probability that at least 5 percent of the sample boxes will be underweight?

SOLUTION

Here the statistic is the sample proportion, the sample size (n = 144) is large enough to assume that the sample proportion is approximately normally distributed with mean

Mean of the sampling distribution of $\hat{p}$

$$\mu_{\hat{p}} = p = 0.10,$$

Standard Error of $\hat{p}$

$$\sigma_{\hat{p}} = \sqrt{\frac{pq}{n}} = \sqrt{\frac{(0.10)(0.90)}{144}}$$

$$= \frac{0.3}{12} = 0.025.$$

22) hyper geometric probability distribution = $\sigma^2 = n\frac{k}{N}\frac{N-k}{N}\frac{N-n}{N-1}$

23) mean = $\mu = n\frac{k}{N}$

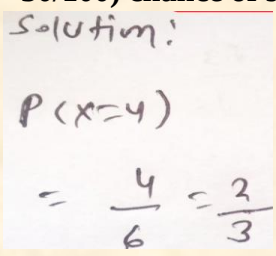
24) Sixty two percent of students get passed in a particular test. This time 500 students appeared in the exam. Using the normal approximation to the binomial distribution, find the mean and variance of this data

Normal approximation to the binomial distribution =

$$\mu = np = 500 \times 0.20 = 100$$
$$\sigma^2 = npq = 500 \times 0.20 \times 0.80 = 80$$
$$\sigma = \sqrt{npq} = \sqrt{80} = 8.94$$

25) Binomial probability distribution = $\lim_{\substack{n\to\infty \\ p\to 0}} b(x;n,p) = \frac{e^{-\mu}\mu^x}{x!}, \quad x = 0,1,2,\ldots,\infty$

26) The incident of occupational disease is such that the workers have 30 percent (P =30/100) chance of suffering from 6 workers. Find P (X=4).



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27) what is hypergeometric experiment?

A hypergeometric experiment is a statistical experiment that has the following properties: A sample of size n is randomly selected without replacement from a population of N items. In the population, k items can be classified as successes, and N - k items can be classified as failures.

28) statement was given and we have to write their null and alternative hypothesis

Formulation of the Null and Alternative Hypotheses: The null hypothesis is "The population mean is 200." The alternative hypothesis is 'The mean is different from 200" or "The mean is not 200." These two hypotheses are written as follows: H0 : $\mu = 200$, H1 : $\mu \neq 200$

29) Standard deviation = $s = \sqrt{\frac{\sum(x-\bar{x})^2}{n-1}}$

30) critical region definition

A critical region, also known as the rejection region, is a set of values for the test statistic for which the null hypothesis is rejected. i.e. if the observed test statistic is in the critical region then we reject the null hypothesis and accept the alternative hypothesis.

31) complete book design and complete Block design mein difference tha..

Randomized complete block designs differ from the completely randomized designs in that the experimental units are grouped into blocks according to known or suspected variation which is isolated by the blocks.

31) 2 table aye ty. 1 mai variance find krni ti or ak mai After(weights) n before(weight) calculate krna ta

Recruit	Weight		Difference, d_i (after minus before)	d_1^2
	Before	After		
1	125	136	11	121
2	195	201	6	36
3	160	158	– 2	4
4	171	184	13	169
5	140	145	5	25
6	201	195	6	36
7	170	175	– 6	25
8	176	190	5	196
9	195	190	14	25
10	139	145	– 5	36
∑	1672	1719	6	673

32) Acceptance region ki definition thi 2 marks

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A confidence interval, also known as the acceptance region, is a set of values for the test statistic for which the null hypothesis is accepted. i.e. if the observed test statistic is in the confidence interval then we accept the null hypothesis and reject the alternative hypothesis.

33) two tail definition

the rejection region is divided equally between the two tails of the sampling distribution of the test-statistic, the test is referred to as a two-tailed test or two-sided test.

34) passion formula

$$P(X=x)=\frac{e^{-\lambda t}(\lambda t)^{x}}{x!},$$

35) LSD formula

$$LSD = t_{\alpha/2,(v)}\sqrt{\frac{2(MSE)}{r}}$$

36) h(y)or g (x) malom karna tha

c) The marginal p.d.f. of X is

$$\begin{aligned} g(x) &= \int_{-\infty}^{\infty} f(x,y)\,dy, && -\infty < x < \infty \\ &= \frac{1}{8}\int_{2}^{4} (6-x-y)\,dy, && 0 \le x \le 2 \\ &= \frac{1}{8}\left[6y - xy - \frac{y^2}{2}\right]_2^4 && 0 \le x \le 2 \\ &= \frac{1}{4}(3-x), && 0 \le x \le 2 \\ &= 0, && x < 0 \ \underline{OR}\ x \ge 2 \end{aligned}$$

Similarly, the marginal p.d.f. of Y is

$$h(y) = \frac{1}{8}\int_{0}^{2} (6-x-y)dx,\ 2 \le y \le 4$$

$$= \frac{1}{4}(5-y), \qquad 2 \le y \le 4$$

$= 0,$ elsewhere.

37) E(X) = $E(x)=\frac{nk}{N}$

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38) Var(X) = $Var(x) = \frac{nk}{N}\left(1-\frac{k}{N}\right)\left(\frac{N-n}{N-1}\right)$

39) Find Critical region : a = 0.05, n = 25 was given

$|Z| < Z_{1-a/2}$
=> a/2 = 0.025
=> 1-a/2 = 0.975
$Z_{1-a/2}$ = 1.96
=>$|Z| < 1.96$

40) T-static = $t = \frac{\bar{x}-\mu_0}{S/\sqrt{n}}$

41) Poisson distribution = $P(X = x) = \frac{e^{-\mu}\mu^x}{x(\text{fact})} \ldots\ldots\ldots(1)$

42) X^2 test statics = $x^2 = \frac{nS^2}{\sigma^2}$

43) Test static = $Z = \frac{X-np_0}{\sqrt{np_0q_0}}$

44)

Question:32.

If n=400, x=275. Then find 95% confidence interval for population proportion.

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We know that,

$$\hat{p} \pm Z_{\alpha/2}\sqrt{\frac{\hat{p}.\hat{q}}{n}} \ldots\ldots\ldots(1)$$

Now,

$$\hat{p} = \frac{x}{n} = \frac{275}{400} = 0.69$$

$$\hat{q} = 1 - \hat{p} = 1 - 0.69 = 0.31$$

$$1 - \alpha = 0.95$$

$$\Rightarrow \alpha = 0.05 \ , \ \alpha / 2 = 0.025$$

putting all values in eq.(1)

$$0.69 \pm Z_{0.025}\sqrt{\frac{(0.69)(0.31)}{400}}$$

$$0.69 \pm Z_{0.025}(0.023)$$

$$0.69 \pm (1.96)(0.023)$$

$$0.69 \pm 0.04508$$

$$\Rightarrow 0.69 + 0.04508 \ , \ 0.69 - 0.04508$$

$$\Rightarrow 0.73508 \ , \ 0.645$$

$$\Rightarrow (0.645, 0.73508)$$

45)

The department claim that the exceeds Rs =.2500 at the 0.05 level , then formulate null alternative hypothesis?

Solution:

Formulation

H_0 :μ < 2500

H_1 :μ > 2500

46) find degree of freedom

v2 = (r-1)(c-1) = (5-1)(4-1) =4 × 3 = 12 degrees of freedom.

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48)

Q4:

Let X be a continuous r.v. with p.d.f.

$$f(x) = 6x(1-x), \qquad 0 \le x \le 1.$$
$$= 0, \qquad \text{otherwise.}$$

Check that $f(x)$ is a p.d.f.

Solution:

The $f(x)$ will be a p.d.f. if

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_0^1 6x(1-x)\,dx = 1$$

$$\int_0^1 6x\,dx - \int_0^1 6x^2\,dx = 1$$

$$\left[\frac{6x^2}{2}\right]_0^1 - \left[\frac{6x^3}{3}\right]_0^1 = 1$$

$$\left[3x^2\right]_0^1 - \left[2x^3\right]_0^1 = 1$$

$$[3(1)] - [2(1)] = 1$$

$$3 - 2 = 1$$

$$1 = 1$$

Hence f(x) is the p.d.f.

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49)

Q5:

Let X and Y be independent r.v's with joint p.d.f.

$$f(x,y) = x^2 + \frac{1}{3}xy, \qquad 0 \le x \le 1, 0 \le y \le 2$$

$$= 0, \qquad \text{elsewhere.}$$

Find the E(X).

Solution:

$$E(X) = \int_x xg(x)\,dx$$

$$g(x) = \int_y f(x,y)\,dy$$

$$= \int_0^2 \frac{3x^2 + xy}{3}\,dy$$

$$= \frac{1}{3}\int_0^2 (3x^2 + xy)\,dy$$

$$= \frac{1}{3}\int_0^2 3x^2\,dy + \frac{1}{3}\int_0^2 xy\,dy$$

$$= \frac{1}{3}\left[3x^2 y\right]_0^2 + \frac{1}{3}\left[\frac{xy^2}{2}\right]_0^2$$

$$= \frac{1}{3}\left[6x^2\right] + \frac{1}{3}\left[2x\right]$$

$$= \frac{6x^2}{3} + \frac{1}{3}\left[2x\right]$$

49)

$$= 2x^2 + \frac{2x}{3}, \qquad for\ 0 \le x \le 1.$$

$$E(X) = \int_x xg(x)dx$$

$$= \int_0^1 x(2x^2 + \frac{2x}{3})dx$$

$$= \int_0^1 2x^3 dx + \int_0^1 \frac{2x^2}{3} dx$$

$$= \left[\frac{2x^4}{4}\right]_0^1 + \frac{2}{3}\left[\frac{x^3}{3}\right]_0^1$$

$$= \left(\frac{2}{4}\right) + \frac{2}{3}\left(\frac{1}{3}\right)$$

$$= \frac{1}{2} + \frac{2}{9} = \frac{9+2}{18} = \frac{11}{18}.$$

50)

Testing of hypothesis about mean of a non-normal population when sample size is large (Unknown Variance)

$$Z = \frac{\bar{x} - u}{S/\sqrt{n}}$$

Testing of hypothesis for the difference between two population means when sample size is large (Unknown Variance)

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (u_1 - u_2)}{\sqrt{\frac{S_1}{n_1} + \frac{S_2}{n_2}}}$$

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$$Z = \frac{\bar{x} - u}{\sigma / \sqrt{n}}$$

Testing of hypothesis for the difference between two population means

For testing of hypothesis of two population means for any sample size and known population variances, we have one formula i.e.

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (u_1 - u_2)}{\sqrt{\frac{\sigma_1}{n_1} + \frac{\sigma_2}{n_2}}}$$

Where:

$\bar{X}_1$ is the mean of the first sample

$\bar{X}_2$ is the mean of the second sample

μ_1 is the mean of the first population.

μ_2 is the mean of the second population

σ_1 is the standard deviation of the population.

σ_2 is the standard deviation of the population.

51)

Questio 41 (Marks: 2)

If there are K treatments and R rows in a Randomized Complete Block Ddesign then calculate the total number of experimental units used.

(k-1)(R-1) = KR-K-R+1

Questio 42 (Marks: 2)

If a car is driven on average no more than 1600 Km per month, then formulate the null and alternative hypothesis.

The Nulll Hypothesis will be

U_o <1600 Km

The Alternative hypothesis will be

U_o >1600 Km

52)

Questio 44 (Marks: 2)

What are the mean and variance of binomial distribution?

Mean = $\mu = np$
Variance = $\sigma^2 = npq$

53)

Construct 95% confidence interval for the difference in means $\mu_1 - \mu_2$ in case of paired observations, where

$\bar{d} = -4.8$, $s_d = 1.32$, $t_{0.025(24)} = 2.0639$

For the above observations the confidence interval will be

$$\overline{X} + t_{\alpha/2(n-1)} \frac{s}{\sqrt{n}}$$

n =25
X =D

54)

Let T_1 and T_2 are two unbiased estimators, variance is

$Var(T_1)=\frac{11}{9}\sigma^2$ *and* $Var(T_2)=\frac{13}{9}\sigma^2$ ***.Which estimator is more efficient and why?***

Solution:

We know that,

$$E_f=\frac{Var(T_2)}{Var(T_1)}=\frac{13\sigma^2/9}{11\sigma^2/9}$$

$$E_f=\frac{13\sigma^2}{9}\times\frac{9}{11\sigma^2}=\frac{13}{11}$$

13/11 is greater than 1.

Which shows that

Var(T_1) < Var(T_2) thus T_2 is more efficient than T_1.

54)

Questio 50 (Marks: 5)

A random sample of 10 university professors gave their salaries (in thousand Rs.) 13, 11, 19, 15, 22, 20, 14, 17, 14, 15.Another random sample of 5 college professors gave their salaries (in thousand Rs.) 9, 12, 8, 10, 16. Calculate the value of pooled estimator S_p.

$$\hat{p}_c=\frac{n_1\hat{p}_1+n_2\hat{p}_2}{n_1+n_2}$$

55)

Questio 52 (Marks: 5)

find n and q.

b) The incident of occupational disease is such that the workers have 30 percent (P =30/100) chance of suffering from 6 workers. Find P (X=4).

Answe

$$P(X=x)=C_x^5\left(\frac{3}{11}\right)^x\left(\frac{4}{11}\right)^{n-x}$$

a) From the given binomial probability distribution find n and q.

b) The incident of occupational disease is such that the workers have 30 percent P =30/100) chance of suffering from 6 workers. Find P (X=4).

A).
n=5
q=4/11
p=3/11

B). P(X=4) =